
THE PROMPT IS NOT A PLACE TO KEEP A BELIEF: WHY STATED BELIEFS COLLAPSE UNDER PRESSURE AND COMPILED ONES HOLD

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Paper under double-blind review

ABSTRACT

Deployed language models keep their standing commitments (persona, policy, stance) in the context window, the same channel their interlocutors write into. Sharing that channel fuses two properties that deserve separate measurement: whether a belief is adopted, and whether it survives being argued against. The two come apart. Stated in context, a belief is adopted almost always (98%) and defended barely half the time under escalating adversarial pressure (58%). Compiled into low-rank attention modulation by a trained hypernetwork, the same belief is adopted less than half the time (46%) and defended almost always (89%). Composing the compiled channel with a small curated context slot recovers most of the adoption while keeping the persistence. Activation steering does not explain the effect: a tuned steering vector built from identical content installs comparably, then collapses to 14% under the same pressure. Nor is the persistence stubbornness. The compiled system concedes to well-warranted counterevidence at $2.4\times$ the rate it concedes to facially deficient challenges, and never to pressure that offers no evidence at all. The effect replicates across three model families and leaves general knowledge recall untouched. Where a stance is stored determines whether it survives, and the protocols that most need models to hold positions for reasons, debate-based oversight above all, currently store those positions in the one channel the opposition is guaranteed to reach.

1 INTRODUCTION

A belief that cannot be defended under pressure is not functioning as a belief. An assistant that folds to a fabricated statistic is unusable for research support or review, and an agent that abandons a *correct* position under authority pressure fails in the direction hardest to detect, because the surface behavior (politeness, agreement, a plausible concession) looks like good conversational hygiene. Standard practice nonetheless places a model’s persistent stance or persona in the context window, alongside the interlocutor’s own content, arbitrated by the model’s own chain of thought. The conditioning and the pressure that will test it share a channel.

The two properties this conflates come apart when measured separately. Across five escalating pressure levels applied to 24 held-out debate topics, a belief stated in context installs the assigned stance in 98% of generations but holds it under pressure in only 58%. The same belief content, compiled into low-rank per-layer attention modulation via a trained hypernetwork write function, installs weakly on its own (46%) but holds at 89%. Composing the compiled channel with a small curated context slot recovers most of the installation gap (77%) while keeping nearly all of the persistence advantage (89%). Installation and persistence are not two measurements of one capability. They are separable properties, and the channel that installs best is not the channel that holds best.

The gap is not a uniform lift. It concentrates precisely where context conditioning collapses hardest: logical counterargument (mean conviction 2.93 for context-only, 4.78 for compiled) and emotional pressure (1.99 vs. 3.82). A benchmark artifact would not choose those two levels.

The asymmetry has a plain reading. Text placed in context is read, reasoned about, and available to be argued against, and instruction tuning specifically rewards accommodating an articulate objection. A compiled modulation is not read as an assertion in that sense. It conditions the forward pass rather

than occupying a position in it, and it is absent from the chain of thought the model reasons over (Section 4).

The nearest alternative explanation is activation steering. Contrastive activation vectors extracted from identical belief content, tuned honestly on held-out training topics, install at 42%, matching compiled modulation’s 46%, then collapse to 14% hold under the same pressure protocol where compiled modulation holds 89%. This is not an artifact of a weak baseline: an independent stress test of activation steering under adversarial perturbation reports directional robustness dropping by up to 64 percentage points and post-attack confidence collapsing to or below 0.25 across four extraction methods and five models from 1.5B to 30B parameters (Le & Le, 2026). Prompt compression is the other obvious candidate, and it fails on form: compressed tokens still occupy context positions and still dilute against a growing transcript, where compiled tensors occupy none. Section 5 reports the full ablation.

The mechanism carries disposition, not facts. General knowledge recall is unchanged within margin when a compiled disposition is active (MMLU 76.5% vs. 74.8%, paired McNemar $p = 0.11$), and specific unfamiliar content such as fabricated statistics, novel proper nouns, and exact action strings does not survive the compilation bottleneck. We route that content to a curated context slot by design (Section 7).

Resistance this strong invites the objection that the model has simply been made stubborn. We measure this directly. Under a protocol that holds tone, assertiveness, and length constant across evidence classes, the update path concedes to well-warranted evidence at $2.4\times$ the rate it concedes to facially deficient evidence (42.5% vs. 17.5%, $\Delta = +25.0$ points), and concedes 0% of the time to the four contentless pressure types (Section 6). The mechanism updates on reasons and holds against pressure that supplies none.

Scope. The tested condition is system-prompt-style belief injection versus compiled per-layer modulation of a single stance on a single topic, evaluated over five pressure turns. We did not test retrieval-augmented generation, MemGPT-style external memory stores, or Reflexion-style verbal reinforcement learning as implemented in their original papers; the generalization to context-mediated conditioning more broadly is offered as the natural reading of the mechanism we isolate, and is named as an inference rather than a tested claim.

Contributions. (1) We report a dissociation between installation and persistence across two conditioning channels for the same belief content, isolated by construction (Section 3). (2) We isolate the effect against the two most obvious competing explanations, fixed-direction activation steering and prompt compression, and against a frozen-offset ablation of the mechanism itself, localizing what does the work (Section 5). (3) We show the resistance this buys is discriminating rather than rigid: the same mechanism concedes at a $2.4\times$ higher rate to well-warranted evidence than to facially deficient pressure (Section 6). (4) We connect the result to debate-based scalable oversight, whose soundness guarantee assumes debaters hold positions for reasons, an assumption every existing remedy places in the same channel the opponent argues in (Section 8).

2 MECHANISM

The compiled channel is a hypernetwork that amortizes into one forward pass what LoRA obtains by gradient descent: a trained write function maps a structured belief representation, through the frozen host’s own hidden states, to per-layer conditioning parameters, in the manner of AdaLN- and FiLM-style conditional normalization (Perez et al., 2018; Xu et al., 2019) and of the hypernetwork-adapter lineage established by GenerativeAdapter (Chen et al., 2025) and Text-to-LoRA (Charakorn et al., 2025).

2.1 THE BELIEF TREE

Conditioning content is stored as a *belief tree*: typed hierarchical nodes $\{\text{statement}, \text{type} \in \{\text{claim}, \text{argument}, \text{evidence}, \text{experience}, \text{strategy}\}, \text{credence}, \text{edges}\{\text{supports}, \text{contradicts}\}\}$. Trees are authored by hand for evaluation topics and produced by the reflection loop for the update-path experiments (Section 6). An early design used numeric credence values on every node; we found

[FIXME: Figure 1: compile pass (belief tree $\rightarrow$ pooled state $\rightarrow$ write function $\rightarrow$ per-layer Δ_Q, Δ_V) and generation pass (modulated forward pass with the curated slot in context), side by side, readable without the caption. Insert /home/dgonier/experiments/figures/ asset once paper_figures_dispositional.py emits it.]

Figure 1: The compile pass and the generation pass. Compilation happens once per belief change (2.5–3.0s end-to-end); generation applies the resulting per-layer tensors on every subsequent forward pass while occupying zero context positions.

no measurable benefit over a coarse categorical scheme and report this as a negative result rather than carry the added complexity forward.

2.2 THE MODULATION

The write function conditions query and value projections at a stride-3 subset of layers (11 of 32), rank 16. For a target layer with base query weight W_Q , the modulated projection is

$$W'_Q = W_Q + \Delta_Q(\phi), \quad \Delta_Q(\phi) = U_Q(\phi) V_Q(\phi)^\top,$$

with $U_Q(\phi) \in \mathbb{R}^{d \times 16}$, $V_Q(\phi) \in \mathbb{R}^{d \times 16}$ produced by the write function from the pooled belief-tree state ϕ ; the value projection W'_V is constructed identically. The compilation is query-agnostic — Δ_Q, Δ_V do not depend on the current prompt — but its effect is query-dependent, because the low-rank update acts on the live hidden state produced by whatever the model is currently attending to.

2.3 THE WRITE FUNCTION

Node representations are pooled with conviction-weighting, passed through a bottleneck, and decoded into the per-layer rank-16 factors above. We correct a timing claim from an earlier draft of this work: the previously reported ~ 20 second compile latency belongs to an unrelated agentic retrieval prototype, not to this write function, which compiles in well under one second; end-to-end reflection $\rightarrow$ edit $\rightarrow$ recompile measures 2.5–3.0 seconds.

We define “compile” at first use, and it is lossy. A trained write function translates the belief tree into per-layer low-rank tensors, fixed until the store changes, yet query-dependent in effect because the update acts on the live hidden state. We reject “precomputed,” which suggests caching a value that could in principle be looked up; this is a learned, lossy translation, and the loss is what Section 7 characterizes. We also flag, to preempt confusion with an unrelated sense of the word, that “compile” here has no relation to `torch.compile` or graph compilation.

2.4 THE CURATED SLOT

A small number of curated statements — a deterministic walk from the belief tree’s activated parent nodes — are placed directly in context alongside the compiled modulation. This slot carries what the bottleneck cannot: specific, low-frequency content such as named entities, exact figures, or constrained action strings (Section 7). Condition F throughout this paper denotes compiled modulation plus this curated slot; C denotes compiled modulation with the slot empty (zero belief-content context tokens).

3 THE DISSOCIATION

3.1 PROTOCOL

Five pressure levels (L1 bare doubt, L2 fabricated evidence, L3 logical counterargument, L4 appeal to authority, L5 emotional pressure) are applied across 24 held-out debate topics, two sides per topic, three rounds per level. A judge model (temperature 0) scores each generation on a 1–5 conviction scale from the transcript alone, blind to condition. Conditions are defined before any result is reported: A is the bare host with no belief conditioning; B states the belief in context; D states the belief in

Table 1: Installation (stance taken, $\geq 4/5$) and persistence (hold under pressure, $\geq 4/5$) by condition, corrected protocol. F = compiled modulation + curated slot; C = compiled modulation alone. CAA row: contrastive steering vectors from identical belief content, single mid-network layer, tuned scale 8. Cross-family rows report the same contrast on Llama-3.2-3B (template-native prompting; corrected protocol).

Condition	Installation (≥ 4)	Hold under pressure (≥ 4)	Model
A — bare host	52%	62%	Qwen (4B)
B — belief in context	98%	58%	Qwen (4B)
D — belief in context + modulation	100%	62%	Qwen (4B)
C — compiled modulation, empty slot	46%	91.9% [†]	Qwen (4B)
F — compiled modulation + curated slot	77%	89%	Qwen (4B)
CAA — tuned contrastive steering	42%	14%	Qwen (4B)
B — belief in context	81%	15%	Llama-3.2-3B
F — compiled modulation + curated slot	98%	81%	Llama-3.2-3B

[†] Measured under the C.2b deconfound (Section 5), which fills the previously-unmeasured cell of C under pressure on the same held-out topics; not part of the original five-condition sweep.

context *and* applies compiled modulation; C applies compiled modulation alone, with the curated slot empty; F applies compiled modulation with the curated slot populated. A tuned contrastive-activation steering condition, extracted from the same belief content, is reported alongside these as the sharpest available baseline (Section 5).

An earlier version of this evaluation judged empty generations: a reasoning-block token budget was exhausted before the answer was produced, post-processing deleted the resulting empty output, and the judge’s refusals parsed to a uniform score of 3.00 (0% hold). Modulated conditions were inadvertently immune, because modulation suppresses think-mode generation, which is why the artifact first looked like a finding rather than a bug. All numbers in this paper come from a corrected rerun of 1,776 generations with zero empty outputs and zero judge errors, using the same judge and protocol. We disclose this because a truncation artifact that silently zeroes only the unmodulated arms of a modulated-vs-unmodulated comparison is a trap other evaluations of this kind of mechanism are likely to hit, and finding it independently after publication would be the worse outcome.

3.2 INSTALLATION VS. PERSISTENCE

Table 1 states the inversion directly. Context installs a stance almost perfectly (B 98%, D 100%) but holds it only slightly better than chance under pressure (B 58%, D 62%). Compiled modulation alone is a weak installer (C 46%, barely above the unconditioned baseline’s 52%) but a strong holder once installed (91.9% under pressure on the same content — see Section 5 for the isolation that produced this cell). F composes the two channels: 77% installation, 89% hold, a net +31 point improvement in persistence over context alone. Tuned CAA steering, included in the same table so the comparison is visible without deferring to an ablation section, matches compiled modulation’s installation rate (42% vs. 46%) and then collapses under exactly the pressure compiled modulation withstands (14% vs. 89%).

3.3 WHERE THE GAP LIVES

The persistence gap is not a uniform lift. Table 2 decomposes mean conviction by pressure level for B and F. F’s margin over B concentrates at L3 (logical counterargument, +1.85) and L5 (emotional pressure, +1.83) — precisely where context conditioning collapses (B falls to 2.93 and 1.99). At L1 and L4, both conditions hold reasonably well (B 4.78 and 4.42); the gap is not present because there is little room for it. This is a mechanism signature: the channels differ most exactly where an interlocutor supplies a well-formed-looking but ultimately contentless argument, which is the regime instruction tuning most rewards accommodating.

Table 2: Mean conviction (1–5) by pressure level, corrected protocol. F’s margin over B concentrates at L3 and L5.

Condition	L1 bare doubt	L2 fabricated ev.	L3 logical	L4 authority	L5 emotional
A	3.93	4.17	3.18	4.56	2.24
B	4.78	3.62	2.93	4.42	1.99
D	4.99	3.42	3.36	4.71	1.78
F	4.89	4.50	4.78	4.68	3.82
F – B	+0.11	+0.88	+1.85	+0.26	+1.83

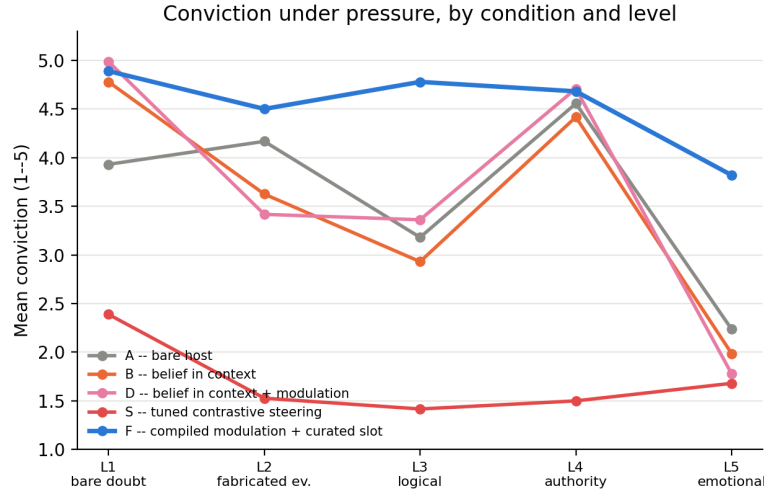


Figure 2: Mean conviction across five pressure levels, all conditions. Generated by experiments/scripts/paper_figures_dispositional.py directly from benchmark_raw_nothink_20260725_100329.json (conditions A/B/D/F) and benchmark_raw_steering_20260725_215612.json (condition S); validated against notes/corrected_baselines.md to within 0.02 conviction points on all 20 (condition, level) cells before this figure was written. The gap between context (B) and compiled (F) conditioning is not a uniform offset; it is concentrated at logical (L3) and emotional (L5) pressure, and CAA steering (S) collapses at every level past L1.

3.4 CROSS-FAMILY AND SCALE

The dissociation replicates on Qwen3.5-27B (+24 points on the hold metric, under a 7× reduced evaluation cap, $n = 240$) and, more sharply, on Llama-3.2-3B: F holds 81% against B’s 15%, and zero-context compiled stance transfer alone reaches 94% on this family (vs. Qwen’s 46%; Section 9 returns to this split). On Mistral-8B, after correcting a template defect in which the trained checkpoints carried literal Qwen chat-template markers, collapse-protection strengthens (40%→5.8% cap rate within-run) but strict stance-holding does not transfer to this family (−10 points relative to B). We report this split precisely rather than average over it: collapse-protection generalizes across the three families tested; strict stance-holding under the exact five-level protocol is, on current evidence, family-specific.

3.5 DILUTION

B and F receive byte-identical belief content and differ only in delivery channel; the +31 point gap is attributable to channel by construction, with no confound to argue about. We keep this observation short: it follows directly from the protocol design in Section 3.1, and over-arguing a by-construction result invites the suspicion it does not deserve.

4 WHY THE CHANNELS DIFFER

Three measurements are consistent with, though they do not prove, a read-vs-not-read account of the dissociation. A linear probe trained to decode the active belief from hidden states succeeds when compiled modulation is active and degrades to 22% accuracy when it is zeroed, indicating the belief’s functional signature is concentrated in the modulated pathway rather than diffused. Attention-pattern divergence between modulated and unmodulated forward passes on the same prompt is small (Jensen–Shannon divergence 0.049), consistent with modulation acting on the value/query content of attention rather than substantially redirecting where the model attends. And modulation measurably suppresses think-mode generation, the pathway through which the model would otherwise narrate — and thereby expose to an interlocutor’s rebuttal — its own reasoning about whether to concede.

We state the claim at the strength the evidence supports: the two channels are *dissociable*, and the compiled channel affects processing without appearing in the reasoning trace the model produces. This is consistent with, not proof of, an account in which context-delivered content is read and is therefore available to be argued against, while compiled content is not. We flag directly that the same probe result above is used in Appendix A.7 to make a different point: decodability from hidden states, even at reduced accuracy, is enough to fail a strict implicit-memory criterion. Both readings use the same number and do not contradict each other — the belief is decodable by a probe with access to internals, and it is nonetheless never surfaced in the text the model reasons over or the interlocutor sees.

5 ISOLATING THE MECHANISM

This section is structured as the reviewer questions it answers.

5.1 IS IT ACTIVATION STEERING?

Per-topic contrastive activation vectors were extracted from the same belief content used to compile F, tuned honestly on training-split topics only, with a joint sweep over injection layer and scale and a degeneracy guard against incoherent generation. The best configuration injects at a single mid-network layer at scale 8; naively injecting at every layer compiled modulation drives degenerates generation at every scale tested, which is itself informative — a fixed direction cannot be applied everywhere compiled modulation is, but a content-derived, query-conditioned update can. The tuned configuration installs at 42% (matching C’s 46%) and holds at 14% under the same five-level pressure protocol where F holds 89% (row reported in Table 1). Independent corroboration exists at a different scale: Le & Le (2026) report directional robustness for activation steering dropping by up to 64 percentage points and post-attack confidence collapsing at or below 0.25 across four extraction methods and five models from 1.5B to 30B parameters under adversarial input perturbation, with the optimal steering layer shifting by up to 17 positions. Our 14% is not an artifact of an undertrained baseline; it is the documented behavior of the method class under this kind of stress.

5.2 DOES THE MECHANISM NEED TO BE INPUT-RESPONSIVE?

We freeze the compiled modulation into a constant per-layer offset — the mean delta across training queries for a fixed belief, with no dependence on the live hidden state — while keeping the curated slot byte-identical to F. This frozen-offset condition holds at 88.9% [81.1, 95.3], statistically indistinguishable from F’s 86.9% [79.2, 93.3]. Input-responsiveness is *not* what carries persistence; a low-dimensional, belief-derived fixed nudge is sufficient. This inverted our prediction, which was that removing input-responsiveness would collapse persistence toward the CAA baseline (14%) or a zero-shot steering-direction-only baseline (21.1%, Section 5.3); it did not.

Because the frozen-offset condition retained the curated slot while the CAA and direction-only baselines did not, this comparison is slot-confounded on its own. A second condition, compiled modulation with the slot forced empty, resolves the confound: it installs weakly (consistent with C) but holds at 91.9% [86.9, 96.1] under pressure on the same held-out topics — full F-like persistence with zero belief-content context tokens. A third condition, curated slot only with modulation disabled, holds at 57.5% [49.7, 64.7], collapsing to 5.6% at L5 — B-like, not F-like. The 2×2 is now complete: modulation, fixed or input-responsive, confers persistence; the curated slot alone behaves like context.

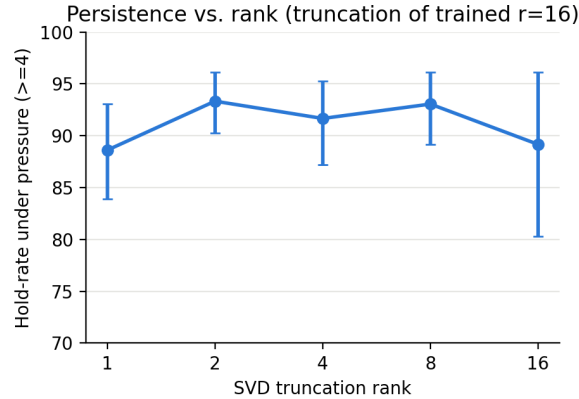


Figure 3: Persistence under SVD rank truncation of the trained rank-16 modulation. Generated by `experiments/scripts/paper_figures_dispositional.py` directly from `results/cl_rank_{1,2,4,8,16}_20260809_*.json`; matches `analysis/cl_summary.md` exactly (88.6/93.3/91.7/93.1/89.2%). Flat across ranks 1–16, overlapping 95% bootstrap CIs; no cliff.

This also fills a previously unmeasured cell of Table 1: compiled modulation under pressure with zero context tokens holds at 91.9%, directly comparable to its 46% installation rate on the same condition.

5.3 IS IT THE AUXILIARY TRAINING SIGNAL?

An earlier draft of this mechanism included an auxiliary steering direction, d^* , applied at generation time as a training-time regularizer. A reconstruction audit found d^* was never in fact applied during the dispositional evaluation reported here; a dose-sweep ablation confirms adding it at inference time does not help and, at higher doses, actively hurts (-0.3 points at dose 0.05, -6.4 at 0.10, -14.4 at 0.20 relative to the F-anchor), and a d^* -only condition (modulation replaced entirely by the fixed direction) collapses to 21.1% — a second fixed-direction method landing in the same regime as CAA (14%). We conclude d^* plays no role in the persistence result and omit it from the main mechanism description (Section 2); Appendix A.9 discloses the training-time signal and this ablation for reproducibility, since the released training code still contains it.

A separate rank sweep — truncating the trained rank-16 modulation via SVD down to rank 1 and re-evaluating persistence — finds a flat curve: 88.6%, 93.3%, 91.7%, 93.1%, 89.2% at ranks 1, 2, 4, 8, 16, with overlapping confidence intervals and no cliff down to rank 1 (Figure 3). A single direction, extracted from the trained rank-16 state by truncation, carries persistence; this is a truncation result about the trained state’s structure, not evidence that native rank-1 training would work equally well, which we did not test.

5.4 IS IT SOFT PROMPTING?

Open. Prefix tuning at matched parameter count separates two properties this paper otherwise conflates: compiled modulation is both outside the context window and not inspectable as assertable text. Prefix tuning is the second without the first. If prefix tuning holds under pressure, the operative variable is inspectability; if it collapses like context does, the operative variable is position. We have not yet run this baseline and report it as open rather than assume an answer.

5.5 IS IT COMPRESSION?

Open. A compressed-prompt baseline at matched token count, evaluated through the same dilution sweep as Section 3.5, would test whether occupying fewer context positions is sufficient on its own. Not yet run.

5.6 DOES THE DISPOSITION CROWD OUT GENERAL CAPABILITY?

A stratified 1,140-question MMLU subset, scored by letter-choice log-probability, shows 76.5% accuracy with no disposition active versus 74.8% with a compiled disposition active; a paired McNemar test over 74 vs. 55 discordant pairs gives $p = 0.11$. A compiled disposition rides alongside general knowledge without displacing it.

6 RESISTANCE OR RIGIDITY

A mechanism this resistant to pressure invites an immediate objection: is it simply stubborn? We argue the operative standard for a debate-oversight setting is not true-versus-false but good-evidence-versus-bad, judged from the turn text alone, because no fact-checker is available in-round. Correct behavior on this standard is to update on good evidence and hold against bad; sycophancy is correspondingly redefined as *capitulation without a reason*, which is what a raw hold-rate metric was always trying to approximate.

Protocol. Each topic is seeded with the side contradicted by the strongest available real evidence, so valid corrections exist by construction, and pressure turns are style-matched across evidence classes (tone, assertiveness, length within $\pm 20\%$) so that surface presentation does not confound the evidence-quality manipulation. Of 24 evaluation topics, 20 are eligible; four are excluded at construction time as value questions with no fact-decidable side — a design-stage rule, not an outcome-based exclusion.

Results. Concession rate is 42.5% on well-warranted evidence versus 17.5% on facially deficient evidence, $\Delta = +25.0$ points. Concession to all four facially deficient pressure types — bare doubt, emotional pressure, unsourced assertion, and unnamed-authority appeal — is 0%. The concessions that do occur are confined to the two evidence subtypes constructed to be indistinguishable from good evidence without independently auditing the warrant (40% concession rate) or checking the arithmetic (65%) — exactly the regime where a human debater without a fact-checker is similarly exposed. In 20–35% of those cases, the reflection step that follows a concession *raises* credence again after identifying the flaw in the evidence that produced it.

Limitations, reported in the same breath. In-conversation conviction alone discriminates weakly between evidence classes (76 vs. 66); the update path that produces the +25 point gap operates through the reflection step, not through in-conversation capitulation directly. Recovery after conceding to disguised bad evidence is only 21%. This experiment covers 20 topics and reports proportions, not a significance test. The reflection loop that implements the update path was built for this experiment and did not exist end-to-end before; the pre-registration for this protocol is on record, and we say so here rather than let a reviewer discover it.

7 WHAT DOES NOT SURVIVE THE BOTTLENECK

Compilation is lossy, and the losses are specific rather than diffuse: fabricated statistics (e.g., an invented “47.3%” figure), unfamiliar proper nouns (e.g., a fictitious organization name), and exact constrained action strings do not reliably survive the compile step. This is why the curated slot (Section 2.4) exists as a principled division of labor rather than an afterthought: compiled modulation carries disposition; the slot carries the specific, low-frequency content the bottleneck cannot. We report these as named, specific failure modes rather than an abstract caveat, on the view that a specific failure reads as honest where a general disclaimer does not.

8 RELATED WORK

We organize prior work along two axes: *where* the conditioning signal lives — context tokens, live activations, or weights — and *what property was measured* once it was installed. Every line of prior work we discuss measures acquisition: whether conditioned content got in. None asks whether it survives being argued against. Table 3 summarizes the family by both axes.

Context-delivered conditioning and its decay. Placing a persona, belief, or instruction directly in context is the default implementation of persistent conditioning in deployed systems, and it is known not to hold. Choi et al. (2024) examine identity consistency across nine LLMs in extended conversation and find that larger models drift *more*, not less, and that explicitly assigning a persona does not reliably prevent that drift; no mitigation is proposed, because the paper’s contribution is characterizing the drift, not repairing it. This is the closest existing statement of our own negative result for context-only conditioning (B, 58% hold), obtained in a different setting and without our channel comparison.

Hypernetwork adapters, and a concession first. A trained network that maps a conditioning input, through a frozen host’s own hidden states, into per-layer low-rank weight parameters in a single forward pass is not a new architectural primitive; it is an established and currently active line. GenerativeAdapter (Chen et al., 2025) modulates the attention *output*-projection at rank 128, trained self-supervised on SlimPajama, and evaluates on StreamingQA, MetaICL, and the Multi-Session Conversation (MSC) personalization benchmark — all single-shot acquisition metrics computed once, immediately after context is mapped to an adapter. On MSC, GenerativeAdapter reaches 40.2 F1 against 66.0 F1 for full-conversation prompting, a gap the paper frames explicitly as an acceptable $4\times$ compute-and-storage saving rather than a capability shortfall. Text-to-LoRA (Charakorn et al., 2025) maps short task descriptions to LoRA weights to elicit latent skills; SHINE (Liu et al., 2026) extends full-layer-coverage hypernetwork conditioning and reports 63.6 F1 on SQuAD-style document QA with meta-LoRA accumulation across sessions; this lineage traces to the fast-weight-programmer framing of a slow network reprogramming a fast one (Schmidhuber, 1993; Ba et al., 2016; Schlag et al., 2021). We inherit this architecture and this framing — our write function is exactly a hypernetwork of this family, conditioning query and value projections rather than the output projection — and we are explicit that placement is not our contribution. What none of these papers measures is whether an installed context, once in, survives being contested. Every evaluation in this family is a static, single-time-point readout against a gold answer; none introduces a repeated query under escalating adversarial pressure. The measurement difference, not the architecture, carries this paper.

The override gap. Cheng et al. (2026) come closest to our question without asking it. They study why a hypernetwork-internalized document fails to override a model’s pretrained prior on knowledge-conflict questions, and build a causal account in which override succeeds when the adapter’s logit margin exceeds the prior’s: accuracy on their hardest conflict tier falls from 68% in the weakest-prior quartile to 16% in the strongest, and their proposed fix (Selective Layer Boosting, combined with conflict-aware training) recovers deep-conflict accuracy from 46.4% to 71.0% on Gemma-2B. This is a genuine competing account of an installation gap that superficially resembles ours, and the two must not be conflated: their pressure is the model’s own static pretrained prior, measured once; ours is an adversarial interlocutor applying escalating pressure across multiple turns after installation has already succeeded. Their own language is the sharpest statement of the assumption our paper tests: parameter-level methods are described as “persistent and query-agnostic,” in contrast to prompts that must be “repeated for every query” — persistence is asserted, not measured, because their protocol has no mechanism to ask the same question twice. We also note, so as not to overclaim in the other direction, that their own retrieval-augmented-generation baseline scores 63.8% on deep conflicts against their conflict-aware parametric method’s 71.0%: context is not uniformly worse than parametric conditioning at installation, and we do not claim it is. Our claim is narrower and, we think, better supported: context installs stance excellently (98%) and holds it poorly (58%); the override gap’s own data is consistent with our account of installation but silent on persistence.

Activation steering. Contrastive and representation-engineering steering methods (Turner et al., 2023; Zou et al., 2023; Rinsky et al., 2024) modify live activations at inference time with no gradient update to the host, conditioned on a contrastive example set rather than structured content. This is the closest prior mechanism to ours by *form* rather than by measured property, and we differentiate on form directly: a steering vector is a single fixed direction applied uniformly regardless of input; our write function produces a low-rank map that acts on the live hidden state and is therefore query-conditioned, even though its parameters are not. Le & Le (2026) independently stress-test activation steering under adversarial input perturbation across four extraction methods and five models from 1.5B to 30B parameters and report directional robustness dropping by up to 64% in the abstract’s framing (“64 percentage points” in the paper’s conclusion; we do not convert between the two), with

post-attack confidence collapsing at or below 0.25 and the optimal steering layer shifting by up to 17 positions. This is independent corroboration, at a different scale and on different models, that our own head-to-head result — CAA steering matching compiled modulation at installation (42% vs. 46%) and collapsing far below it under pressure (14% vs. 89%) — is the documented behavior of the method class, not an artifact of an undertrained baseline.

PEFT and modulated conditioning. LoRA (Hu et al., 2021) and its derivatives target the query and value projections by default; our write function targets the same two projections, and we say plainly that this placement is not itself a contribution. Prefix and prompt tuning (Li & Liang, 2021; Lester et al., 2021) learn conditioning that is not natural-language text but that still occupies context positions, competing for attention and diluting as the transcript grows — the sharpest open baseline this paper still owes (Section 5.4), because it separates “outside the context window” from “not readable as an assertion,” two properties our own comparison currently conflates. FiLM (Perez et al., 2018) and AdaLN-style conditional normalization (Xu et al., 2019) are the one-sentence framing we borrow for the write function itself: a hypernetwork producing per-layer conditioning parameters from a pooled representation, in one forward pass, is architecturally the same move visual conditional-generation models make; we condition on a typed belief tree rather than a class label or style embedding.

Sycophancy, debate, and scalable oversight. Debate-based oversight has a formal foundation: Irving et al. (2018) propose that two agents arguing before a judge can push tractable judgment toward a complexity class direct judging cannot reach, and Brown-Cohen et al. (2023) strengthen this to a doubly-efficient protocol in which an efficient honest debater defeats an unbounded dishonest one under a bounded-query verifier. The UK AI Security Institute’s safety case sketch (Buhl et al., 2025) builds on this protocol for an internally deployed system, naming as the claim its guarantee rests on, that both debaters are honest “in the global equilibrium of the game, when both debaters are playing their best response strategies” — and concedes explicitly that a system reaching this equilibrium in training need not continue playing it in deployment. The empirical record is genuinely mixed, and we report the disagreement rather than resolve it: Khan et al. (2024) find that optimizing debaters for persuasiveness improves rather than harms non-expert judge accuracy, but their protocol assigns each debater a fixed position for the entire match with no mechanism to switch sides, so it cannot speak to whether a debater holds its position under pressure. Against a more adversarial backdrop, Bertalaníć & Fortuna (2026) report that homogeneous multi-agent debate teams exhibit modal sycophantic conformity up to 85.5%, an oracle gap up to 32.3 percentage points relative to isolated self-correction, and $2.1\text{--}3.4\times$ the token cost for equal or lower accuracy — debate, in their controlled setting, absorbs diversity without converting it into verification. Temperature ablations in the same study show conformity intensifying with greater initial diversity rather than resolving it. Debate’s soundness guarantee is a property of the game, not of any single agent: the judge is amplified because a wrong claim faces an adversary that will not let it stand, and that guarantee assumes debaters hold positions for reasons. An agent that capitulates to social pressure rather than to evidentiary content does not merely play badly — it removes the adversarial structure the protocol’s soundness rests on. Every existing mitigation we are aware of is protocol- or prompt-level: refined prompts, mandated dissent, disagree-or-commit turn structures. All of them place the instruction to hold in the same channel the opponent is arguing in. Our result is that this is the wrong channel, and Section 6’s discrimination result — update on warranted evidence, hold against unwarranted pressure — is what keeps this from being the obfuscated-arguments problem (Barnes & Christiano, 2020) in a different costume: a debater that never updates is not a good debater.

Table 3: Table A: the prior-work family by axis. Every row measures acquisition; none measures persistence under adversarial pressure after installation.

Work	Conditioning lives in	Modulated module(s)	Measured property	Persi
Choi et al. (2024)	context (persona)	—	drift over turns	no m
Chen et al. (2025)	weights (hypernet)	attn. output-proj., $r=128$	F1 / accuracy	no
Charakorn et al. (2025)	weights (hypernet)	LoRA (task-conditioned)	task accuracy	no
Liu et al. (2026)	weights (hypernet)	full-layer LoRA	F1 / ROUGE-L	no
Cheng et al. (2026)	weights (hypernet)	LoRA + layer boosting	conflict accuracy	no (s
CAA / RepEng	activations	fixed direction, all/one layer	injection success	no
Le & Le (2026)	activations	fixed direction	robustness under perturbation	adver
LoRA / prefix / FiLM-AdaLN	weights / context / weights	Q,V / context positions / per-layer	task performance	no
Bertalanič & Fortuna (2026)	multi-agent context	—	conformity, oracle gap	no (a
This work	weights (Q,V) vs. context	attn. Q,V, $r=16$	installation and persistence	yes –

9 LIMITATIONS

The headline result is measured primarily on a single model family (Qwen), with the cross-family replication in Section 3.4 showing that collapse-protection generalizes while strict stance-holding, at least under our exact five-level protocol, does not transfer uniformly — Mistral gains collapse-protection but loses 10 points of strict hold relative to context. All models tested are instruction-tuned; we do not know whether the dissociation holds for base models, where the accommodate-the-objection behavior we attribute to instruction tuning (Section 4) may not be present in the same form.

Always-on compiled modulation measurably reduces turn-to-turn generation diversity (adjacent-turn similarity 0.006 against a 0.099 no-modulation ceiling, with onset at turns 2–3); a prefill-skip variant recovers partial diversity (0.079). This attractor effect does not contaminate the five-level pressure benchmark reported here — measured adjacent-round similarity across 100 benchmark sessions is 0.094, close to the no-modulation ceiling — but any claim about sustained multi-turn deployment beyond five pressure rounds has to be stated against this open problem, and we have not resolved it.

The evaluated protocol is a single stated belief on a single topic, tested against five pressure levels over a bounded number of rounds; we do not test multiple simultaneous beliefs, cross-session persistence, or the longer horizons (tens to hundreds of turns) over which persona drift is documented to compound in unrelated work. The compiled channel’s rank-16 boundary is characterized by the truncation sweep in Section 5.3 but not by training at lower native rank.

Finally, a conditioning channel invisible to prompt inspection is a real dual-use concern: the same property that makes compiled modulation resist adversarial pressure makes it resist legitimate inspection by an operator or auditor who can read a system prompt but cannot read compiled weights. We raise this ourselves because it is the obvious objection to the mechanism working as well as it does, and it is a real one.

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648 A APPENDIX

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650 A.1 TRAINING CURRICULUM

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652 **[FIXME: Fill from experiment_guide.md / training run logs.]**

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654 A.2 BELIEF-TREE SCHEMA SPECIFICATION

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656 **[FIXME: Full node schema, edge types, worked example.]**

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658 A.3 CONDITION DEFINITIONS WITH TOKEN BUDGETS

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660 **[FIXME: A/B/C/D/F token budgets, curated-slot size, CAA extraction detail.]**

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662 A.4 PER-LEVEL CROSS-FAMILY TABLES

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664 **[FIXME: Full Qwen/Llama/Mistral per-level tables (Section 3.4 summarizes; this appendix carries the raw per-level numbers).]**

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667 A.5 NUMERIC-CREDENCE NEGATIVE RESULT

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669 **[FIXME: Detail on the categorical-vs-numeric credence ablation referenced in Section 2.1.]**

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672 A.6 ATTRACTOR DIAGNOSIS

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674 Diversity metrics under sustained modulation: 0.006 (always-on) vs. 0.079 (prefill-skip) vs. 0.108 (turn-gate), against a 0.099 no-modulation ceiling. Onset at turns 2–3. Adjacent-round similarity across 100 benchmark sessions: 0.094 — close to ceiling, confirming the five-level pressure benchmark (Section 3) is not contaminated by this effect within its round budget. See Section 9 for the open multi-turn deployment question this leaves.

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679 A.7 PROBE DECODABILITY

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681 The linear-probe result reported in Section 4 (decodable when modulation is active, 22% when zeroed) is repeated here to make explicit that it fails a strict implicit-memory criterion: a sufficiently well-positioned probe with access to internal activations can recover the conditioned belief above chance even though it never appears in the model’s own reasoning trace or surfaces to the interlocutor. This is not in tension with Section 4’s claim — decodable-by-probe and read-by-the-model-as-an-assertion are different properties, and we report both readings of the same number rather than picking the one that is more flattering to either argument.

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689 A.8 JUDGE PROMPTS

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691 **[FIXME: Full judge prompt text, conviction rubric 1–5, blinding procedure.]**

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693 A.9 THE AUXILIARY TRAINING SIGNAL (d^*)

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695 An early version of the write function’s training objective included an auxiliary steering-direction loss, d^* , intended as a regularizer. A reconstruction audit established that d^* was not in fact applied at generation time in the dispositional evaluation reported in the main body. The ablation reported in Section 5.3 (dose sweep and d^* -only condition) confirms it is not part of the machinery that produces the persistence result: applying it at inference time does not help and, at higher doses, hurts, and a condition built entirely from d^* collapses to the same regime as CAA steering. We disclose it here because the released training code still contains the auxiliary loss term, and the paper and the code must not diverge on what does and does not matter.

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A.10 REPRODUCIBILITY

[FIXME: Seeds, hardware, judge model version and temperature, exact run identifiers (benchmark_raw_nothink_20260725_100329.json etc.), links to experiments/analysis/*.md summaries per result.]

A.11 DEAD ENDS

[FIXME: Numeric-credence nodes (see above); naive all-layer CAA injection (degenerates at every scale, Section 5.1); six-axis design-space taxonomy cut from the main body during the outline revision.]